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Recommended Problem Statements

Internal Smart India Hackathon (SIH) 2026

Sardar Patel Institute of Technology, Mumbai · August 2026

Part A — Software Domains (18 picks)

1. Artificial Intelligence & Machine Learning

Title	**Uncertainty-Aware Diabetic Retinopathy Triage for Primary Health Centres**
Detailed Description	India has over 100 million diabetics, yet retinopathy screening barely reaches rural patients because ophthalmologists are concentrated in cities; a 2025 Scientific Reports study of Indian primary healthcare found AI-assisted community screening is the most scalable model but faces implementation barriers. Students must build an offline-first triage tool for Primary Health Centres using low-cost smartphone fundus-camera images: an image-quality gate that rejects ungradable captures with retake guidance, a DR severity grader trained on Indian datasets (e.g., APTOS 2019 from Aravind Eye Hospital), calibrated uncertainty estimates so ambiguous cases are flagged for human review instead of silently misclassified, and a referral-priority queue for district hospitals. Key challenges: domain shift across cameras, class imbalance for severe grades, model compression for edge devices, and clinician-trustable explanations (lesion heatmaps).
Expected Solution / Outcome	A deployable edge application (TFLite/ONNX, <50MB) demonstrating quality-gating, five-grade DR classification with quadratic-kappa >0.85 on APTOS validation data, calibrated referable/non-referable triage with abstention, Grad-CAM lesion overlays, and a teleophthalmology referral dashboard. Impact: enables a single technician-run PHC camp to safely screen hundreds of patients per day, prioritising sight-threatening cases.

2. Generative AI & Large Language Models

Title	**Voice-First Vernacular Scheme Navigator with Verifiable, Citation-Grounded Eligibility Reasoning**
Detailed Description	Hundreds of central and state welfare schemes go under-utilised because beneficiaries — often low-literacy, non-English speakers — cannot discover eligibility or complete applications; GoI's push to make portals like e-Shram available in all 22 scheduled languages via Bhashini shows the scale of the language-access problem. Students must build a voice-first agent (not a text chatbot) that conducts a spoken dialogue in Indic languages using ASR/TTS (Bhashini APIs or open Indic models like IndicWhisper), runs retrieval-augmented generation over official scheme documents, executes deterministic eligibility rule-checking (age, income, occupation, state) via LLM function-calling rather than free-form generation, and returns every answer with verifiable citations to the source paragraph. Key challenges: code-mixed speech, hallucination suppression on high-stakes benefits information, structured slot-filling over noisy voice input, and latency on low-bandwidth connections.
Expected Solution / Outcome	A demo where a user speaks in Hindi/Marathi (plus one more language), is interviewed conversationally, and receives a ranked list of schemes they qualify for, each with cited source clauses and a pre-filled application summary. Measured deliverables: citation-faithfulness above 90% on a test question set, end-to-end voice latency under 5 seconds, and a documented zero-hallucination eligibility engine.
Reasoning & India Relevance	Sourced from the PIB release on Bhashini-enabling e-Shram in 22 languages — proof that vernacular access to entitlements is a live GoI priority — and AWS's Indic voice AI

	work (Polly bilingual Hindi/Indian-English) showing industry investment in Indian-language speech stacks. Innovation lies in separating generative dialogue from deterministic, citation-verified eligibility logic, directly attacking LLM hallucination where it harms citizens most. India affinity is maximal: Bhashini, DPI, welfare delivery, 22 scheduled languages.

3. Cybersecurity & Digital Forensics

Title	**Real-Time Guardian Against Digital Arrest and Deepfake Video-Call Scams**
Detailed Description	India lost over Rs 22,495 crore to cybercrime in 2025, and 'digital arrest' scams—where fraudsters impersonate CBI, police, or customs officials on video calls, often using deepfaked faces and spoofed IVR calls—disproportionately target senior citizens. Victims stay isolated on calls for hours while coerced into transferring savings. Students must build an on-device Android companion that, with user consent, monitors incoming call/video sessions and flags scam signals in real time: deepfake and face-liveness artifacts, synthetic-voice markers, scripted keywords ('arrest warrant', 'skip verification'), caller-ID spoofing patterns, and screen-share coercion, then triggers a trusted-contact alert and one-tap 1930 helpline/I4C reporting with an auto-captured forensic evidence bundle. Key challenges: low-latency on-device inference, multilingual (Hindi + regional) speech analysis, and privacy-preserving processing.
Expected Solution / Outcome	A working Android app plus lightweight ML pipeline that demonstrates live detection on staged scam calls: deepfake/liveness score, scam-script keyword alerts in at least two Indian languages, a guardian-alert SMS flow, and an exportable, hash-sealed evidence package (audio snippets, transcript, metadata) formatted for the National Cybercrime Reporting Portal. Demo target: detection within 60 seconds of scam-call onset with measurable precision on a curated test set.

4. Data Science & Big Data Analytics

Title	**Graph-Powered Mule Ring Detection for Streaming UPI-Scale Transactions**
Detailed Description	Government data reported to Parliament shows UPI fraud crossed Rs 805 crore across 10.6 lakh incidents in FY26 alone, with low recovery rates; fraudsters increasingly launder through networks of rented 'mule' accounts that per-transaction rule engines cannot see. AWS's ML blog demonstrates that graph neural networks over accounts, devices and merchants outperform tabular models for exactly this pattern. Students must build a streaming analytics pipeline: a realistic synthetic UPI transaction generator (VPAs, devices, timestamps, fraud typologies like cashback scams and mule chains), ingestion via Kafka/Kinesis-style streams, incremental construction of a heterogeneous transaction graph, GNN or community-detection scoring to surface coordinated mule rings within minutes, and an investigator console with explainable subgraph evidence. Key challenges: class imbalance, near-real-time graph updates at high throughput, and explainability for regulator-grade alerts.
Expected Solution / Outcome	An end-to-end demo processing thousands of synthetic transactions per second, detecting seeded mule rings with precision/recall reported against baseline rules (target >20-point F1 improvement), median detection latency under one minute, and visual subgraph explanations per alert. Impact: a blueprint banks and fintechs can adapt to curb India's fastest-growing digital payment fraud vector.

5. Healthcare Technology

Title	**Sahayika: Offline-First Voice Copilot for ASHA Workers' Home Visits**
Detailed Description	India's one million ASHA workers are the backbone of rural primary healthcare, yet they receive only about 23 days of basic training, work where doctors are distant and mobile signal is unreliable, and spend hours hand-filling multiple paper registers after every home visit. Students must build a voice-first, Indic-language mobile copilot that works fully offline: it answers protocol questions (danger signs in pregnancy, immunization schedules) from an on-device or cached knowledge base, lets the ASHA dictate visit findings conversationally, auto-structures them into the standard government reporting formats, and syncs to an ABDM-style backend when connectivity returns. Key challenges include low-resource Indic speech recognition on budget Android phones, offline-online sync with conflict resolution, guardrailed medical Q&A that escalates uncertain cases, and a UI usable by low-digital-literacy users.
Expected Solution / Outcome	A working Android app plus sync server demonstrating an end-to-end visit: spoken Hindi/Marathi dialogue captured offline, auto-generated digital register entries, a supervisor dashboard, and ABHA-linked record push on reconnection. Measurable impact: cut per-visit documentation time from 15-20 minutes to under 5 and reduce transcription errors, demonstrated live with sample visit scripts.

6. Smart Education

Title	**NIPUN Mitra: Indic Speech AI for ASER-Style Oral Reading Fluency Assessment**
Detailed Description	ASER 2024 national findings show large shares of rural Grade 3 and Grade 5 children cannot read grade-level text, yet fluency today is measured through slow, subjective one-on-one assessments by surveyors or teachers—impossible to run frequently at classroom scale. Industry has proven the alternative: Seesaw built an AI reading assessment on Amazon Transcribe for English. Students must build the Indic-language equivalent for India's NIPUN Bharat foundational literacy mission: a teacher's app that records a child reading a Hindi or Marathi passage, runs Indic ASR tuned for child speech, computes words-correct-per-minute, detects mispronunciations, substitutions and hesitations, maps performance onto ASER reading levels (letter/word/paragraph/story), and tracks each child's trajectory. Key challenges: child-speech ASR in Indic languages, noisy-classroom robustness, alignment-based miscue detection, and offline/low-bandwidth operation.
Expected Solution / Outcome	A mobile app plus scoring engine demoed live: a child's recorded Hindi reading scored within seconds, per-word error highlighting, automatic ASER-level classification validated against human raters on a small test corpus, and a class dashboard flagging children needing remediation. Impact: per-child assessment time cut from minutes of teacher attention to a self-serve 60-second recording.

7. FinTech & Digital Payments

Title	**Voice-First Vernacular UPI with Anti-Coercion Safeguards for Feature-Phone Users**
Detailed Description	India has hundreds of millions of feature-phone and low-literacy users who remain outside the UPI revolution. NPCI's UPI 123Pay (IVR, missed-call, sound-based payments) exists, but field studies report sluggish adoption: voice recognition fails on regional pronunciations, there is no payee memory, and assisted onboarding exposes vulnerable users to fraud. Students must build a voice-first payment assistant—deployable as an IVR flow plus low-end Android app—that completes UPI-style transactions entirely through conversational speech in at least three Indian languages. It must remember frequent payees, confirm amounts by reading them back in the user's language, and add safeguards 123Pay lacks: voice-biometric authentication to stop assisted-onboarding fraud, and anomaly prompts when a transaction deviates from the user's pattern. Challenges: robust Indic ASR on noisy calls, dialect handling, spoof-resistant voice auth, and a simulated UPI switch backend.
Expected Solution / Outcome	A demonstrable end-to-end prototype: phone-call IVR (or app-simulated IVR) completing person-to-merchant payments via voice in three languages against a mock UPI backend, with payee favorites, spoken confirmation, and voice-ID verification that rejects a second speaker in live demo. Measurable targets: task completion under 90 seconds, ASR intent accuracy benchmarked per language, and a documented fraud-safeguard test matrix.

8. Agriculture & Precision Farming

Title	**Satellite-Driven Monsoon Sowing and Irrigation Advisor for Smallholder Plots**
Detailed Description	Over 85% of Indian farmers cultivate under two hectares, and most depend on increasingly erratic monsoons to time sowing and irrigation. Commercial precision-agriculture tools assume large fields, drone budgets, and reliable connectivity, so smallholders get no plot-level guidance. Students must build a software platform that fuses free open satellite data (Sentinel-2 NDVI/soil-moisture proxies), IMD gridded rainfall forecasts, and the farmer's plot boundary (drawn on a map or imported from AgriStack-style land records) to generate weekly sow/irrigate/hold advisories per plot. Key challenges: computing vegetation and moisture indices for irregular sub-hectare boundaries despite cloud cover during monsoon, calibrating advice against local crop calendars, delivering it in regional languages over WhatsApp/SMS for low-bandwidth users, and quantifying advisory confidence so farmers can trust or discount it.
Expected Solution / Outcome	A working web plus WhatsApp/SMS prototype where a demo farmer draws a plot and receives a vernacular weekly advisory card (sow/irrigate/hold with confidence score) generated from real Sentinel-2 and rainfall data for at least two Maharashtra districts. Measurable impact: plot-level guidance at near-zero marginal cost, targeting 10-20% irrigation water savings and reduced resowing losses from failed monsoon onset.

9. Smart Cities & Urban Mobility

Title	Green-Corridor Adaptive Signal Timing from Low-Cost Camera and Probe Data
Detailed Description	The TomTom Traffic Index 2025 ranks Bengaluru, Kolkata, Pune, and Mumbai among the world's slowest cities, with commuters losing about 168 hours a year to gridlock; most Indian signals still run fixed timers tuned years ago. Full adaptive-signal hardware is unaffordable at scale. Students must build a software-only system that estimates queue lengths and turning movements from existing junction CCTV feeds (or recorded Indian traffic video) using computer vision robust to India's heterogeneous traffic — two-wheelers, autos, buses, pedestrians sharing lanes — fuses this with probe/GPS speed data, and computes optimized signal plans plus emergency-vehicle green corridors via reinforcement learning or heuristic optimization, validated in SUMO simulation of a real corridor. Key challenges: vehicle detection in dense mixed traffic, sim-to-real calibration, and explainable timing recommendations traffic police can trust.
Expected Solution / Outcome	A dashboard for a modeled Indian corridor (e.g., a Mumbai arterial) showing live queue estimation from video, recommended cycle plans, and simulated before/after results demonstrating 15-25% reduction in average junction delay in SUMO, plus a one-click green-corridor mode for ambulances. Deliverables: vision pipeline, optimization engine, simulation validation report, and integration notes for city ICCC deployment.

10. IoT & Edge Computing

Title	Edge Analytics for Jal Jeevan Rural Water Networks on 2G-Class Links
Detailed Description	Jal Jeevan Mission has deployed sensor-based IoT devices to monitor rural drinking water supply — flow, pressure, chlorine, and pump status — but village schemes sit behind 2G-class links, and raw high-frequency telemetry either floods the network or arrives too late to catch leaks, dry-running pumps, and contamination events. Students must build an edge analytics layer for the village pump house: on-device stream processing that detects leak signatures (night-flow analysis), pump dry-run and motor anomalies, and water-quality excursions locally; adaptive sampling and delta-encoding that cuts uplink volume by 90%+; and an offline-capable operator panel for the village waterman (jal mitra) in local language. Key challenges: designing bandwidth-aware telemetry protocols (MQTT with compression/batching), edge anomaly models that run on microcontroller/Pi hardware, prioritized alerting when connectivity is intermittent, and district dashboards reconstructing network state from sparse summaries.
Expected Solution / Outcome	A prototype with simulated or real sensor streams for a model village scheme: edge node demonstrating leak and dry-run detection within minutes, measured 90%+ uplink data reduction versus raw streaming on a throttled link, offline jal-mitra panel, and a district dashboard ranking schemes by non-revenue water and downtime. Deliverable includes a replayable event dataset so judges can verify detection latency.

11. Cloud Computing & DevOps

Title	**Chaos-Engineering Harness for UPI-Scale Digital Public Infrastructure Resilience**
Detailed Description	UPI processes billions of transactions monthly, yet repeated 2025 outages — analyzed by ORF as a systemic resilience gap — showed how bank-side slowdowns and year-end rushes cascade into national payment failures, eroding trust in India's DPI. Most teams test happy paths; few can rehearse partner-bank latency spikes, dependency timeouts, or 10x festival-day surges. Students must build an open chaos-engineering and resilience-scoring harness for a reference UPI-like microservice stack (switch, PSP, and bank simulators on Kubernetes): declarative fault injection (latency, error storms, pod kills, network partitions — patterned on AWS Fault Injection Simulator concepts), synthetic Diwali-scale load generation, automatic verification of degradation behaviors (circuit breakers, retries with backoff, queue shedding, idempotent reconciliation of stuck transactions), and a resilience scorecard per service. Key challenges: realistic failure taxonomies from documented outages, blast-radius control, and reconciliation correctness under partial failure.
Expected Solution / Outcome	A deployable testbed: reference payment stack, chaos controller with 8-10 codified experiments replaying documented UPI failure modes, load generator, and an automated resilience report scoring recovery time, failed-transaction reconciliation accuracy, and user-visible error rate. Demo shows the baseline stack failing a bank-latency experiment, then passing after applying recommended circuit-breaker and queue-shedding patterns.

12. Blockchain & Web3

Title	**Programmable Digital Rupee Sandbox for Leak-Proof Welfare and Subsidy Delivery**
Detailed Description	RBI's retail e-rupee pilot is testing programmability—money that a sponsor (government or corporate) can constrain to designated purposes, merchant categories, geographies, and expiry dates—with Direct Benefit Transfer as a headline use case, plus offline functionality for low-connectivity areas. Yet no open sandbox exists for experimenting with welfare programmability policies before they touch real money. Students must build a CBDC-style sandbox: token engine with purpose-bound money implemented via smart contracts, a government console to issue programmed DBT vouchers (e.g., fertilizer subsidy spendable only at registered agri-merchants before season end, unspent balance auto-returned), merchant and beneficiary wallets, and an offline-transfer mode using signed tokens exchanged over Bluetooth/QR with double-spend reconciliation on reconnect. Challenges: enforcing spend conditions on-chain, offline double-spend prevention, privacy of beneficiaries versus auditability, and interoperable redemption to regular money.
Expected Solution / Outcome	A deployable sandbox demonstrating an end-to-end subsidy cycle on a testnet: government issues 100 programmed vouchers, beneficiaries spend at compliant vs non-compliant mock merchants (non-compliant blocked on-chain), two phones complete an offline QR/Bluetooth transfer later reconciled without double-spend, and an audit dashboard proves zero leakage with full anonymized traceability. Includes a policy-simulation report quantifying leakage prevented versus an unconstrained-transfer baseline.

13. E-Governance & Digital Public Services

Title	**Voice-First Multilingual Grievance Copilot for CPGRAMS with Smart Triage and Deduplication**
Detailed Description	CPGRAMS, India's central grievance portal run by DARPG, receives lakhs of complaints monthly, yet citizens must navigate text-heavy forms, guess the right ministry among thousands of subordinate offices, and decode opaque status updates — a barrier for non-English speakers and first-time users. IIT Kanpur's IGMS 2.0 proved AI can analyze grievances for officials, but no citizen-side voice interface exists. Students must build a voice-first assistant that accepts spoken grievances in Indic languages (Bhashini-style ASR and translation), uses retrieval-augmented classification to route complaints to the correct ministry/office, detects duplicate or repeat grievances via text embeddings, scores urgency, and returns plain-language status summaries. Key challenges: code-mixed speech recognition, hallucination-safe routing with explainable justifications, and low-resource language handling.
Expected Solution / Outcome	A working web/WhatsApp-style demo where a citizen speaks a grievance in an Indian language and receives a correctly routed, deduplicated, tracked complaint with a plain-language acknowledgment. Measurable impact: reduced misrouting and re-lodging, grievance filing time under three minutes for non-English speakers, and an official-side triage dashboard showing urgency clusters.

14. Environmental Sustainability

Title	**Satellite Farm-Fire Detection with Field-Level Attribution and Farmer Advisory Nudges**
Detailed Description	Stubble burning in Punjab and Haryana is a major driver of North India's winter smog, but official fire counts rely on thermal detections from polar-orbiting satellites whose fixed overpass times miss fires lit outside those windows — research shows systematic underreporting — making enforcement reactive and advisory outreach poorly targeted. Punjab now uses satellite data to counsel farmers, creating demand for better tooling. Students must build a pipeline on open Sentinel-2 imagery that detects burned areas post-facto through spectral change detection even when active-fire passes miss the event, attributes burns to village/field polygons, predicts high-risk villages from harvest-timing signals, and pushes Punjabi-language advisories on residue-management alternatives and subsidies. Key challenges: cloud and haze masking, burned-area classification with limited labels, temporal differencing at scale, and a usable enforcement-versus-advisory dashboard.
Expected Solution / Outcome	A geospatial web application showing detected burn scars for one district-season with village-level attribution, an accuracy comparison against published fire-count baselines, a predicted risk ranking of villages for the coming week, and sample multilingual advisory messages. Impact: recovering fires missed by overpass-time gaps and shifting response from post-facto penalty to pre-emptive farmer engagement.

15. Disaster Management

Title	**Ward-Level Urban Flood Nowcasting and Geo-Targeted Alerts with Verified Crowdsourced Reports**
Detailed Description	Mumbai, Chennai, and Bengaluru flood almost every monsoon, yet warnings remain city-wide and generic: NDMA's SACHET/CAP system can broadcast alerts to phones, but there is no street-level intelligence telling a commuter which underpass is waterlogged right now or which ward will flood in the next two hours. Students must build a system that fuses IMD rainfall observations and forecasts, open elevation data identifying low-lying pockets and drainage chokepoints, and crowdsourced geotagged waterlogging reports verified through computer-vision checks on submitted photos, to nowcast ward-level flood risk and publish CAP-compatible, geo-targeted alerts in local languages with safe-route suggestions. Key challenges: filtering spam and false reports, nowcasting with sparse sensor ground truth, alert-fatigue thresholds, and graceful degradation when connectivity fails during storms.
Expected Solution / Outcome	A demo replaying a historical Mumbai or Chennai flood event showing ward-level risk evolving hourly, citizen photo-reports being auto-verified and fused into the risk map, and geo-targeted CAP-format alerts with safe routes issued only to affected wards. Impact: street-level actionable warnings versus today's city-wide advisories, with a false-report rejection rate metric.

16. Tourism & Culture

Title	**Predictive Crowd-Safety and Darshan Slot Orchestration for Pilgrimage Sites and Festivals**
Detailed Description	Recurring stampedes — Maha Kumbh 2025 and repeated temple tragedies such as in Andhra Pradesh — show crowd management at India's religious gatherings remains reactive; AI command centers exist only at mega-events, while thousands of mid-sized temples handling festival surges have nothing. Students must build an affordable crowd-safety stack: crowd-density estimation from CCTV or drone frames using counting CNNs; short-horizon inflow forecasting from slot bookings, transit signals, and festival-calendar history; dynamic darshan slot booking with QR-paced entry; automated alerts when corridors approach unsafe person-per-square-metre thresholds, with re-routing suggestions for barricade and signage teams; and a multilingual pilgrim app showing live wait times. Key challenges: density estimation in dense occluded crowds, running inference on cheap edge hardware, and building a crowd-flow simulator to demonstrate interventions safely.
Expected Solution / Outcome	A prototype that processes recorded festival crowd footage to produce live density heatmaps and unsafe-threshold alerts, a slot-booking flow that demonstrably flattens simulated arrival peaks, and an administrator console replaying a surge scenario with recommended re-routing. Impact: a deployable sub-lakh-rupee alternative to bespoke mega-event command centers for mid-sized pilgrimage sites.

17. Supply Chain & Logistics

Title	**DIGIPIN-Powered Address Intelligence to Cut Failed Deliveries in India's Last Mile**
Detailed Description	Indian addresses are unstructured and landmark-based ('behind the temple, near the water tank'), causing failed and repeated delivery attempts that inflate last-mile costs — the most expensive leg of e-commerce logistics, which AWS's Dynamic Delivery Planner work shows is highly optimizable. India Post's DIGIPIN now assigns every 4m x 4m square a geocode, but tooling to connect messy real-world addresses to DIGIPINs barely exists. Students must build an address-intelligence stack: an ML parser that converts free-text, code-mixed Hindi-English addresses into structured components and predicts the DIGIPIN with a confidence score; deduplication that clusters variant spellings of the same household; a rider app with DIGIPIN navigation; and a feedback loop where confirmed deliveries continuously refine geocodes, exposed as an open API for small ONDC-style logistics providers. Key challenges: NER on code-mixed text, ambiguity resolution, and outcome-driven geocode learning.
Expected Solution / Outcome	A demo where a batch of realistic messy Indian addresses is parsed, deduplicated, and geocoded to DIGIPINs with accuracy measured against ground truth, plus a rider app simulation showing delivery confirmation improving subsequent geocode confidence. Impact: measurable reduction in simulated failed-delivery attempts and route distance, with an open API small logistics players can consume.

18. Accessibility & Assistive Technologies

Title	**ISL Setu: Two-Way Indian Sign Language Avatar Interpreter for Public Service Counters**
Detailed Description	India has an estimated 18 million Deaf citizens but only a few hundred trained sign language interpreters, so routine interactions at hospital registration desks, banks, railway counters and government offices routinely happen with no communication support at all. AWS has demonstrated the core pattern with GenASL—generative AI converting speech/text into expressive ASL avatar animations—but nothing equivalent exists for Indian Sign Language. Students must build a counter-top web/tablet app for two-way conversation: staff speech (Hindi/English) is transcribed, translated into ISL gloss, and rendered as a 3D avatar signing from a library of ISL animations; the Deaf user replies via sign captured on camera and classified into text, or via an ISL-icon quick-response board. Key challenges: ISL gloss grammar (which differs from spoken word order), assembling and smoothing sign animation sequences, and real-time webcam sign recognition for a bounded service-domain vocabulary.
Expected Solution / Outcome	A working kiosk-style demo handling a complete scripted transaction (e.g., hospital OPD registration) both directions: live speech rendered as an ISL avatar with under 3-second latency, and at least 30-50 domain signs recognized from webcam input with reported accuracy. Deliverables include the reusable ISL animation dictionary and gloss-translation module.

Part B — Hardware Domains (15 picks)

19. Embedded Systems

Title	RailNetra: Edge-AI Embedded Trolley for Detecting Rail Cracks and Missing Fasteners
Detailed Description	Rail fracture is documented as the leading cause of train derailments on Indian Railways, which still relies heavily on keymen walking thousands of route-kilometres doing visual and hammer checks. Students must build a low-cost, battery-powered embedded unit mountable on a push-trolley that combines a camera, IMU, GPS and vibration sensing on a Raspberry Pi/ESP32, running a quantized edge-vision model to flag surface cracks, missing fastener clips and joint gaps in real time, geotagging each defect and syncing to a maintenance dashboard when connectivity returns. Key technical challenges: reliable on-device inference under vibration and variable daylight, fusing accelerometer signatures with vision to cut false positives, sub-metre geotagging, rugged enclosure design, and multi-hour battery operation trackside.
Expected Solution / Outcome	A working trolley prototype demonstrated on a mock track section that detects seeded defects (cracks, missing clips) with high precision, geotags them onto a live maintenance dashboard, and runs for hours on battery. Impact: converts manual keyman patrolling into data-backed preventive maintenance across 68,000+ route-km, directly attacking the top cause of derailments.

20. Internet of Things (IoT)

Title	ColdKavach: LoRaWAN Cold-Chain Guardian Predicting Shelf-Life of India's Perishable Produce
Detailed Description	The Government's NABCONS study across 54 crops (2020-22) confirms India loses a large share of fruits and vegetables after harvest, with broken or unmonitored cold chains between farm gate, mandi, reefer truck and retail a core cause. Students must build battery-operated LoRaWAN sensor pucks (temperature, humidity, door-open, shock/tilt) that ride inside crates and reefers, a low-cost gateway (Raspberry Pi plus LoRa concentrator), and an edge rules/ML engine that converts cumulative time-temperature exposure into a remaining-shelf-life estimate per crate, raising SMS alerts before spoilage thresholds; a 2-8°C vaccine mode addresses immunisation logistics. Key challenges: multi-month battery life, LoRa link budget from inside metal reefer bodies, calibrated low-temperature sensing, offline data buffering, and a shelf-life model light enough to run on the gateway.
Expected Solution / Outcome	An end-to-end demo of three sensor pucks, one gateway and a live dashboard showing cold-chain excursions flagged within a minute, per-crate shelf-life countdowns, and SMS alerts reaching a farmer/FPO phone. Impact: a low-cost weapon against post-harvest losses that GoI's NABCONS study values in the tens of thousands of crores annually.

21. Robotics & Automation

Title	**SafaiMitra: Ultra-Low-Cost Manhole Cleaning Robot to End Human Entry in Sewers**
Detailed Description	Nearly 500 Indian sanitation workers have died cleaning sewers and septic tanks since 2019 despite the legal ban, and while robots like Genrobotics' Bandicoot are being deployed in Rajasthan and metro cities, their price keeps them out of reach of smaller urban local bodies. Students must build a portable, collapsible manhole robot: a frame-mounted robotic arm with a grabber bucket lowered into the manhole, waterproof camera and lighting streaming to an operator console, an onboard gas-sensing (H_2S, CH_4, CO) safety interlock, and edge vision that classifies blockage type (silt, plastic, construction debris) to guide the operator. Key challenges: corrosion- and water-proofing to IP65+, high-torque arm design on a student budget, tether and power management, and a teleoperation interface usable by municipal sanitation staff.
Expected Solution / Outcome	A prototype demonstrated on a mock manhole that grabs and lifts 2-5 kg of simulated sludge, streams live video with gas readings to the operator console, and targets a component cost under Rs 50,000 — an order of magnitude below commercial robots — making mechanised cleaning viable for small municipalities under NAMASTE funding.

22. Drones & UAV Applications

Title	**Drone Didi Sahayak: Variable-Rate Spraying and Crop-Stress Payload for SHG Agri-Drones**
Detailed Description	The Rs 1,261-crore Namo Drone Didi scheme is equipping 14,500 women self-help groups with agricultural drones for spraying fertilisers and pesticides, but today's spraying is largely uniform-rate — wasting chemicals — and Drone Didis lack tools to prove service quality to farmer customers. Students must build a retrofit smart payload: a PWM-controlled pump and nozzle module fused with a downward RGB/NIR camera on ESP32/Raspberry Pi that maps crop stress (NDVI-proxy) in a first survey pass, then modulates spray rate zone-wise on the spraying pass, while logging per-acre coverage, wind-based drift conditions and chemical consumption into an auto-generated service report. Key challenges: staying within tight payload weight budgets, real-time geofenced variable-rate actuation, robust vision under harsh field lighting, and DGCA-aligned flight and spray logging.
Expected Solution / Outcome	A demonstrable payload (bench rig plus small-drone or gimbal demo) achieving 20-30% simulated chemical reduction via zone-wise variable-rate spraying, and an auto-generated Hindi service report for the farmer. Impact: higher earnings and safety for the 14,500-strong SHG drone fleet while cutting pesticide overuse and input costs for smallholders.

23. Electric Vehicles & Smart Mobility

Title	**Retrofit Edge-AI Battery Guardian with Thermal-Runaway Early Warning for Electric Two-Wheelers**

Detailed Description	India's EV boom runs on two- and three-wheelers, but a wave of scooter battery fires forced BIS to notify new performance and safety standards (IS 17855) for lithium-ion traction packs. Budget e-2Ws use minimal BMS hardware that cuts off charging but cannot anticipate thermal runaway. Students must build a low-cost retrofit battery guardian node: an ESP32 with multi-point NTC temperature probes, pack current and voltage sensing, a VOC/hydrogen off-gas sensor and an IMU, clipped onto a demonstration Li-ion pack. A TinyML model running on-device learns normal charge-discharge signatures and flags failure precursors — abnormal temperature-rise rate, cell-imbalance drift, off-gassing — raising a local siren and a cloud alert. Key challenges: a safe fault-injection test rig, multi-sensor fusion, sub-second on-device inference, and ultra-low BOM cost.
Expected Solution / Outcome	A working retrofit safety pod demonstrated on a lab battery pack with safely simulated fault signatures, showing early-warning alerts well before cutoff thresholds, a quantified false-alarm rate, a digital-twin style dashboard of pack health, and a bill of materials under Rs 2,000 — making predictive battery safety affordable for India's mass-market electric two-wheelers.

24. Renewable Energy Systems

Title	**Intelligent PM-KUSUM Solar Pump Controller with Dry-Run, Theft and Irrigation Smarts**
Detailed Description	PM-KUSUM Component B targets 20 lakh standalone solar agricultural pumps, but pumps in remote fields fail from dry-running as water tables drop, panels get stolen, and farmers irrigate whenever the sun shines rather than when crops need water, wasting groundwater. Students must build a retrofit intelligent pump controller: an ESP32 performing motor current signature analysis to classify dry-run, cavitation and bearing faults at the edge; tilt and geofence sensors with LoRa/GSM alarms for panel theft; and irrigation scheduling that fuses soil-moisture and irradiance data, sending vernacular SMS advisories without internet connectivity. Key challenges: fault classification from current waveforms on a microcontroller, off-grid power budgeting, rugged low-cost packaging, and farmer-friendly interaction design for non-technical users.
Expected Solution / Outcome	A bench prototype with a small DC pump and panel demonstrating automatic dry-run shutdown within seconds, a panel-theft alarm, and soil-moisture-aware scheduling that measurably reduces pumped water versus sun-only operation — a sub-Rs 3,000 retrofit that protects farmer assets, motor life and groundwater under PM-KUSUM.

25. Smart Healthcare Devices

Title	**ABDM-Ready Multi-Vital Screening Kit for ASHA Workers at Rural PHCs**
Detailed Description	India's rural primary health centres face chronic shortages of doctors and diagnostics, and frontline ASHA/ANM workers still record vitals on paper, so early signs of hypertension, anaemia and infection are missed. Students must build a rugged, battery-powered screening kit — ESP32/Raspberry Pi with low-cost SpO2, single-lead ECG, temperature, blood-pressure and heart-rate sensors — that captures multiple vitals in one guided workflow, runs on-device TinyML triage models to flag at-risk patients fully offline, and syncs FHIR-formatted records to Ayushman Bharat Digital Mission (ABDM) health accounts when connectivity returns. Key challenges: medical-grade signal conditioning on cheap sensors, offline-first store-and-forward

	architecture, on-device inference within microcontroller limits, multilingual voice guidance for semi-literate operators, and tamper-evident data logging.
Expected Solution / Outcome	A handheld prototype that screens five vitals in under three minutes, flags high-risk readings offline via an on-device model, and demonstrates end-to-end sync of an ABDM-compatible FHIR record. Impact: one ASHA worker can reliably triage 50+ villagers per day with a device costing under Rs 5,000, shrinking referral delays at understaffed PHCs.

26. Wearable Technologies

Title	**Heat-Stress Early-Warning Wearable for India's Outdoor Workers**
Detailed Description	India's construction labourers, farm and MGNREGA workers, and gig delivery riders face intensifying heatwaves that cause thousands of excess deaths and hit the poorest hardest. Heat illness is preventable if rising physiological strain is caught early, but weather-app warnings ignore individual physiology and worksite micro-climate. Students must build a low-cost armband or helmet-clip wearable — ESP32 with skin-temperature, PPG heart-rate, humidity and activity sensors — running an on-device model that fuses personal strain indicators with WBGT-style micro-climate readings to issue graded rest-and-hydration alerts through vibration, plus a supervisor dashboard aggregating workers over LoRa/BLE. Key challenges: motion-artifact-tolerant PPG on cheap sensors, personalised baselines, sweat- and dust-proof enclosures, week-long battery life, and alerting that works for workers without smartphones.
Expected Solution / Outcome	A wearable prototype under Rs 1,500 demonstrating live physiological-strain scoring, graded vibration alerts, and a site-supervisor dashboard aggregating five or more workers over LoRa/BLE, validated in a simulated exertion demo. Impact: early detection of heat strain before symptomatic collapse, supporting state Heat Action Plans for the workforce most exposed to climate risk.

27. Smart Agriculture Systems

Title	**Solar Edge-Vision Pest Scout with Vernacular Voice Advisories for Smallholders**
Detailed Description	Indian smallholders lose 15-25% of crop value to pests and diseases and often spray pesticides on calendar schedules rather than evidence, because agronomists rarely reach small plots and connectivity-dark fields make cloud-only vision tools useless. NASSCOM's agritech analysis identifies sensor- and vision-driven decision support as the key digitisation gap, and AWS's connected-farm work proves camera-plus-sensor fusion — but at large-farm price points. Students must build a solar-powered field stake: Raspberry Pi or ESP32-CAM imaging a pheromone trap, with soil-moisture and leaf-wetness sensors, running quantised on-device models that count and classify trap catches, spot leaf-disease signatures, and trigger irrigation or spray advisories delivered as vernacular voice calls or SMS. Key challenges: TinyML vision under a one-watt solar budget, Indian pest-trap dataset curation, disease-versus-nutrient-deficiency confusion, and LoRa mesh relay to a connected gateway.
Expected Solution / Outcome	A field-stake prototype under Rs 7,000 that autonomously photographs a pest trap, counts and classifies catches on-device, fuses soil-moisture data, and issues a vernacular voice or SMS advisory — demonstrated end-to-end with no internet at the node. Impact: evidence-based spray decisions that cut pesticide cost and residue for smallholder plots.

28. Industrial Automation

Title	**GPU-Free Edge Vision Inspection Station for Zero-Defect MSME Production Lines**
Detailed Description	India's ZED (Zero Defect Zero Effect) certification drive is pushing lakhs of MSMEs toward defect-free manufacturing, yet manual visual inspection on small production lines is fatiguing, inconsistent and misses micro-defects, while GPU-based machine vision systems are unaffordable. Students must build a low-cost inline visual inspection station: a Raspberry Pi camera rig with controlled LED lighting over a moving demo conveyor, running quantized anomaly-detection and defect-localization models on CPU or a low-cost accelerator at the edge, with a servo reject actuator and defect heatmap logging. Synthetic defect images may augment scarce training data, mirroring AWS generative-AI practice for manufacturing. Challenges: real-time CPU-only inference, lighting invariance, training with under 100 defect samples, and actuator synchronization at line speed.
Expected Solution / Outcome	A tabletop conveyor demo inspecting parts such as fasteners, PCBs or castings at one or more parts per second, localizing defects with a heatmap, auto-rejecting faulty items, and producing a ZED-ready quality log — an under-Rs-12,000 station that replaces fatigued manual inspection on MSME lines.

29. Smart Sensors & Instrumentation

Title	**DoodhParakh: Handheld Multi-Sensor Milk Adulteration Analyser for Village Dairy Collection Centres**
Detailed Description	Government releases document continuing FSSAI enforcement against milk adulterants such as urea, detergent, maltodextrin and added water, and national surveys keep milk quality a live public concern — yet village collection centres typically test only fat/SNF, with lab confirmation taking days. Students must build a handheld analyser combining multi-wavelength LED photometry/NIR reflectance, electrical conductivity, pH and temperature sensing in a small flow cell, with an on-device ML classifier (ESP32/Raspberry Pi) trained on deliberately spiked samples to flag common adulterants and estimate water dilution within 60 seconds, logging QR-stamped results against each farmer's ID for transparent payment. Key challenges: optical repeatability in turbid milk, cross-sensor feature fusion, calibration transfer between devices, rapid cleaning between samples, and building a credible spiked-sample training dataset.
Expected Solution / Outcome	A benchtop-validated handheld unit distinguishing pure milk from water-diluted and urea/detergent/starch-spiked samples with high accuracy in demo, 60-second test time, a sub-Rs-6,000 bill of materials, and a tamper-evident digital quality log per farmer. Impact: fairer farmer payments and consumer safety across India's hundreds of thousands of dairy collection points.

30. Energy Management Systems

Title	**Feeder-Edge Sentinel: Distribution Transformer Health and Theft-Localization Node for DISCOMs**

Detailed Description	RDSS requires smart metering of all feeders and distribution transformers to cut India's chronic AT and C losses, yet Prayas Energy Group reports that by March 2025 only about 3 percent of sanctioned DT meters were actually communicating, leaving DISCOMs blind to overload and theft. Students must build a clamp-on feeder-edge sentinel for distribution transformers: split-core CTs and voltage sensing on three phases, oil and ambient temperature probes, and an ESP32 computing load imbalance, a transformer thermal-aging index, and an energy-balance audit comparing DT input against aggregated downstream consumption to localize theft pockets — with alerts over GSM/LoRa. Challenges: accurate polyphase energy measurement with hobby-grade sensors, energy-balance algorithms robust to meter gaps, harsh-environment packaging, and ultra-low-bandwidth telemetry.
Expected Solution / Outcome	A lab microgrid demonstration with a toy transformer and multiple metered loads, where the node detects an induced unmetered theft tap and an overload event, publishes a transformer health index, and renders a DISCOM-style analytics dashboard — a sub-Rs 4,000 device addressing the DT-metering gap national programs have struggled to fill.

31. Defence & Security Technologies

Title	**Low-Cost Acoustic-RF Drone Intrusion Alert Grid for Border Villages**
Detailed Description	Hostile small drones ferrying narcotics and arms across the India-Pakistan border have surged, with hundreds of intrusions along Punjab alone; Carnegie and ORF analyses show military-grade counter-UAS systems are too scarce and costly to blanket thousands of kilometres of border and vulnerable installations. Students must build a strictly passive, detection-only node network: microphone arrays capturing rotor-acoustic signatures plus RF scanning of common 2.4/5.8 GHz control links on ESP32/Raspberry Pi, with on-device classifiers distinguishing drones from birds, motorbikes and generators. Nodes fuse over LoRa to estimate approximate bearing and push alerts to a control-room dashboard and village volunteers. Key challenges: acoustic classification at low outdoor SNR, false-positive suppression, multi-node time synchronisation for bearing estimation, and all-night solar-battery operation. No jamming or interception — early warning only.
Expected Solution / Outcome	A three-node demonstration grid, each node under Rs 6,000, that detects a hobby quadcopter at 100+ metres, classifies it against confusers, estimates bearing via node fusion, and raises dashboard and SMS alerts within seconds. Impact: affordable persistent early warning that complements scarce high-end counter-UAS systems along smuggling-prone border stretches.

32. Assistive Devices for Persons with Disabilities

Title	**ADIP-Priceable Smart Cane for Indian Streets: Obstacles, Buses, Signage**
Detailed Description	Policy analysis citing the WHO/UNICEF Global Report shows about 90% of people needing assistive technology lack access, and imported smart mobility aids cost tens of thousands of rupees — far beyond the subsidy norms of India's ADIP scheme for Divyangjan. Indian streets also pose hazards Western canes never modelled: head-height overhangs, open drains, and chaotic bus stops where visually impaired commuters cannot identify approaching route numbers. Students must build a clip-on smart cane module: ultrasonic/ToF sensors for ground-level and head-height obstacles with learnable haptic codes, plus an ESP32-CAM or Raspberry Pi vision unit that reads

	bus route numbers and multilingual signage aloud using on-device OCR and text-to-speech. Key challenges: reliable ranging in rain and dust, low-latency haptic encoding, OCR of Indic-script signage in motion, full-day battery, and a bill of materials within ADIP subsidy ceilings.
Expected Solution / Outcome	A clip-on cane module under Rs 4,000 demonstrating overhang and drop-off detection with distinct haptic codes, bus route-number announcement at a mock bus stop, and Hindi/English signage reading — evaluated through a blindfolded obstacle-course trial. Impact: an assistive aid whose cost fits ADIP subsidy limits, enabling government-scheme mass distribution rather than boutique sales.

33. Smart Infrastructure & Building Automation

Title	**Solar-Powered Clip-On Structural Health Nodes for India's Ageing Bridges**
Detailed Description	India recorded about 170 bridge collapses and 202 deaths between 2021 and 2025, with experts blaming poor inspection and absent modern monitoring rather than funding; continuous structural health monitoring exists only on marquee bridges. Students must build a solar-powered clip-on SHM node for ageing bridges and footbridges: high-resolution MEMS accelerometers capturing ambient vibration, strain gauges and tilt sensing on ESP32 or Raspberry Pi, computing modal frequencies at the edge and flagging drift in natural frequency or damping that indicates stiffness loss, plus traffic overload event counting and LoRa alerts to authorities. A lab bridge model with removable members demonstrates damage detection. Challenges: extracting modal features from noisy ambient data on-device, temperature compensation, energy harvesting for years of unattended operation, and alert thresholds that avoid false panic.
Expected Solution / Outcome	A working node network on a model bridge that detects induced damage such as a loosened member through a measurable modal-frequency shift, logs overload events, and pushes early-warning alerts to a dashboard — a sub-Rs 5,000-per-node kit enabling mass instrumentation of India's thousands of ageing district bridges and footbridges.